Example 1.1

Write the decimal forms of the following rational numbers:

i) $\frac{1}{4}$ ii) $1\frac{3}{20}$ iii) $-5\frac{4}{5}$ iv) 3 v) $\frac{1}{3}$

Solution:

(i) $\frac{1}{4}=\frac{25}{100}=0.25$ (ii) $1\frac{3}{20}=\frac{23}{20}=\frac{115}{100}=1.15$ (iii) $-5\frac{4}{5}=\frac{-29}{5}=\frac{-58}{10}=-5.8$

(iv) $3=\frac{3}{1}=\frac{30}{10}=3.0$ (v) $\frac{1}{3}=0.3333...$ (by actual division and it is recurring and non-terminating)

Note

- The above examples show how a rational number may be given in decimal form. The reverse process of converting the decimal form of a rational number to the fractional form may be seen in the higher classes.
- There are decimal numbers which are non-terminating and non-recurring such as

  π = 3.14159265358979323846264 3.....

  $\sqrt{2}$ = 1.41421356237309504880168..... etc.

They are not rational numbers and one can study more about them in the higher classes.

Try these

Write the decimal forms of the following rational numbers:

1. $\frac{4}{5}$ 2. $\frac{6}{25}$ 3. $\frac{486}{1000}$ 4. $\frac{1}{9}$ 5. $3\frac{1}{4}$ 6. $-2\frac{3}{5}$

1.2.3 Positive and Negative rational numbers

Rational numbers may be classified as positive and negative rational numbers.

If both the numerator and the denominator of the fraction representing a rational number are of the same sign, then the rational number is positive.

For example, numbers like $\frac{3}{4}, \frac{-11}{-6}$ etc., are positive rational numbers.

If either the numerator or the denominator of the fraction representing a rational number is negative, then the rational number is negative.

For example, numbers like $\frac{-3}{4}, \frac{11}{-6}$ etc., are negative rational numbers.

Note

- 0 is a rational number which is neither positive nor negative.
- Note that $\frac{-11}{6}=\frac{11}{-6}=-\frac{11}{6}$

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